

Q What is folic acid and why do I need to take it?

Folic acid is a B-group vitamin that is key to the formation of your baby's spine, brain, and nervous system. These organs are some of the very first to develop, which is why you are encouraged to take folic acid even before you conceive.

The neural tube is the part of the embryo that develops into your baby's brain and spinal chord. Making sure you have good levels of folic acid in your system throughout preconception, conception, and the first three months of pregnancy will help minimize the risk that your baby will develop neural tube defects by 72 percent. In dads-to-be, good folate levels can reduce the instance of sperm abnormalities.

Essential supplement

Folic acid is the synthesized version of vitamin B₉. When it occurs naturally in food, it is known as folate and is present in leafy green vegetables such as cabbage. Studies show that our bodies are better at using the synthesized version so look for supplements that contain 5-methyltetrahydrofolic acid, which is already "biologically active." When you are hoping to conceive, ideally both of you should start taking a daily supplement of 400mcg folic acid before you stop contraception. When pregnant, keep taking it until you are at least 12 weeks pregnant. If you are considered high risk (if either of you has a medical or family history of a neural tube defect; if a previous pregnancy was affected; if you are over 35; or if you are diabetic or epileptic), you may be advised to take a higher dosage of 5mg daily.

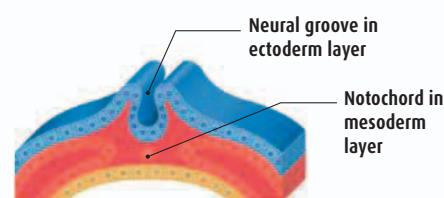


Somites run in pairs along the neural tube.

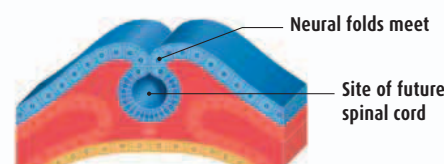
Neural tube By week five of pregnancy, there is a neural tube running down the middle of the embryo, complete with somites that will become vertebrae.

NEURAL TUBE FORMATION

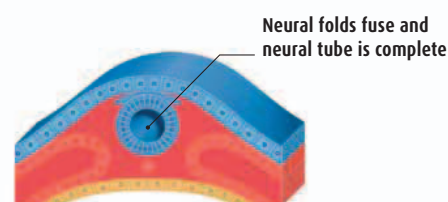
In the earliest stages, your baby forms from three primary layers of cells. These layers fold to create the basic structures of the body, including the important neural tube.



1 A neural groove begins to form when the top layer (ectoderm) sinks toward a column of cells called the notochord in the middle layer (mesoderm).



2 As the groove deepens, its edges come together to make a tube shape. This tube extends along the embryo's back, eventually forming the spine and brain.



3 Folic acid helps the folds of the tube to fuse. If the tube doesn't close fully, it can result in birth defects such as spina bifida.

Whether you're already pregnant or trying to get pregnant, start taking **folic acid** as soon as you can.

Q We are trying for a baby. Do we need to change what we eat or drink?

If you lead a healthy lifestyle and eat a good, balanced diet, you've already established the right pattern. If not, make positive changes to your nutrition now to help the health of your sperm and eggs (which each take three months to develop) and establish the necessary reserves of nutrients for a healthy pregnancy. Also eliminate, or reduce to within guideline levels, alcohol and caffeine, which have a detrimental effect on conception. Eat more phytoestrogens—found in linseed, whole wheat, and lentils—to balance hormones in both partners, and eat a colorful range of antioxidant-rich foods to boost the quality and motility of sperm.

Q Is it true that what you eat can affect how long it takes to conceive? We have fertility issues.

Anything—including diet—that can affect hormones and the health of the woman's eggs, the man's sperm, the fallopian tubes, and the uterus can have an impact on how long it takes to get pregnant. So yes, your diet can adversely affect your internal chemistry.

Your weight—whether you're overweight or underweight—is also an important factor if there are fertility issues, since it can determine whether or not you have too little or too much estrogen to ovulate. For a man, being underweight can affect the quality of his sperm. Eat the right balance of healthy, nutritious,

unrefined foods from the main food groups, avoid the "empty calories" of junk food, and have three regular meals a day to help you regulate your hormones and boost your chances of conception in the months ahead.

Some women may be advised to take a **prenatal multivitamin** in addition to folic acid, but in general you should be able to get all the **nutrients** you need from your diet.